

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1 Load required packages
library(tidyverse)
library(rvest)
library(lubridate)
library(here)

# Check working directory
here()
```

```
## [1] "/home/guest/EDEfall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2 Read the Durham 2024 LWSP web page into a webpage object
the_url <-
  "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
the_website <- read_html(the_url)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3 Scrape system information and maximum daily use data

# Water system name
the_system_name <- the_website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
print(the_system_name)
```

```
## [1] "Durham"
```

```
# PWSID
the_pwsid <- the_website %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()
print(the_pwsid)
```

```
## [1] "03-32-010"
```

```
# Ownership
the_ownership <- the_website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
print(the_ownership)
```

```
## [1] "Municipality"
```

```
# Maximum Day Use (MGD)
max_day_use <- the_website %>%
  html_nodes("th~ td+ td") %>%
  html_text()
print(max_day_use)
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

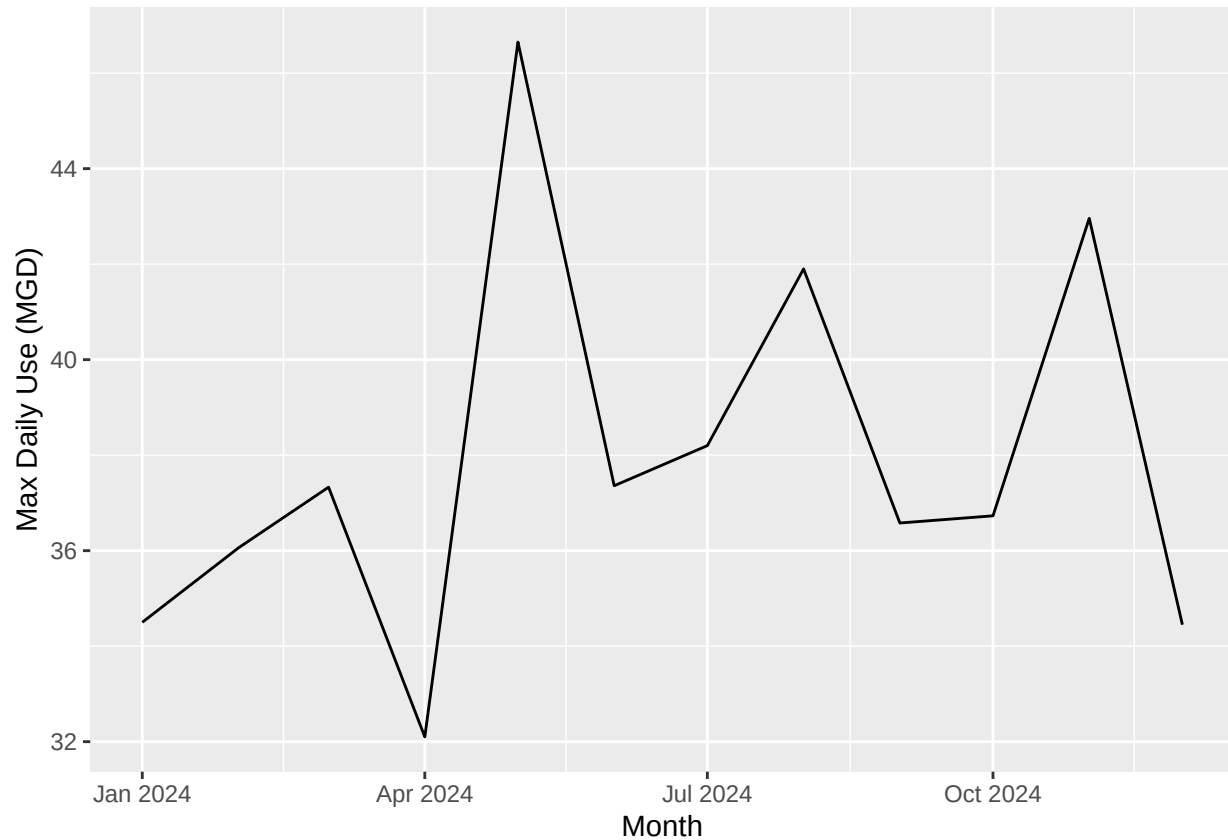
5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4 Create dataframe
# Create month order
month_order <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                 "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

df_lwsp <- data.frame(
  "Month" = month_order,
  "Year" = rep(2024, 12),
  "Max_Day_Use_MGD" = as.numeric(max_day_use),
  "System_Name" = rep(the_system_name, 12),
  "PWSID" = rep(the_pwsid, 12),
  "Ownership" = rep(the_ownership, 12)
) %>%
  mutate(Date = my(paste(Month, Year)))
df_lwsp
```

##	Month	Year	Max_Day_Use_MGD	System_Name	PWSID	Ownership	Date
## 1	Jan	2024	34.50	Durham	03-32-010	Municipality	2024-01-01
## 2	Feb	2024	36.06	Durham	03-32-010	Municipality	2024-02-01
## 3	Mar	2024	37.33	Durham	03-32-010	Municipality	2024-03-01
## 4	Apr	2024	32.10	Durham	03-32-010	Municipality	2024-04-01
## 5	May	2024	46.65	Durham	03-32-010	Municipality	2024-05-01
## 6	Jun	2024	37.36	Durham	03-32-010	Municipality	2024-06-01
## 7	Jul	2024	38.20	Durham	03-32-010	Municipality	2024-07-01
## 8	Aug	2024	41.90	Durham	03-32-010	Municipality	2024-08-01
## 9	Sep	2024	36.58	Durham	03-32-010	Municipality	2024-09-01
## 10	Oct	2024	36.73	Durham	03-32-010	Municipality	2024-10-01
## 11	Nov	2024	42.96	Durham	03-32-010	Municipality	2024-11-01
## 12	Dec	2024	34.45	Durham	03-32-010	Municipality	2024-12-01

```
#5 Plot maximum daily withdrawals across the months for 2024
ggplot(df_lwsp, aes(x = Date, y = Max_Day_Use_MGD)) +
  geom_line() +
  labs(x = "Month",
       y = "Max Daily Use (MGD)")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6. Create a scraping function for LWSP data
scrape_lwsp <- function(pwsid, year) {

  # Construct the URL
  the_url <- paste0(
    "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=",
    pwsid,
    "&year=",
    year
  )
}
```

```

)

# Read the website
the_website <- read_html(the_url)

# Scrape system information
the_system_name <- the_website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()

the_pwsid <- the_website %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()

the_ownership <- the_website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()

# Scrape 12-month maximum daily use (MGD)
max_day_use <- the_website %>%
  html_nodes("th~ td+ td") %>%
  html_text()

# Create month order
month_order <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                 "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

# Build dataframe
df_lwsp <- data.frame(
  "Month" = month_order,
  "Year" = rep(year, 12),
  "Max_Day_Use_MGD" = as.numeric(max_day_use),
  "System_Name" = rep(the_system_name, 12),
  "PWSID" = rep(the_pwsid, 12),
  "Ownership" = rep(the_ownership, 12)
) %>%
  mutate(Date = my(paste(Month, Year)))

# Return dataframe
return(df_lwsp)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7 Fetch Durham 2020 data using our function

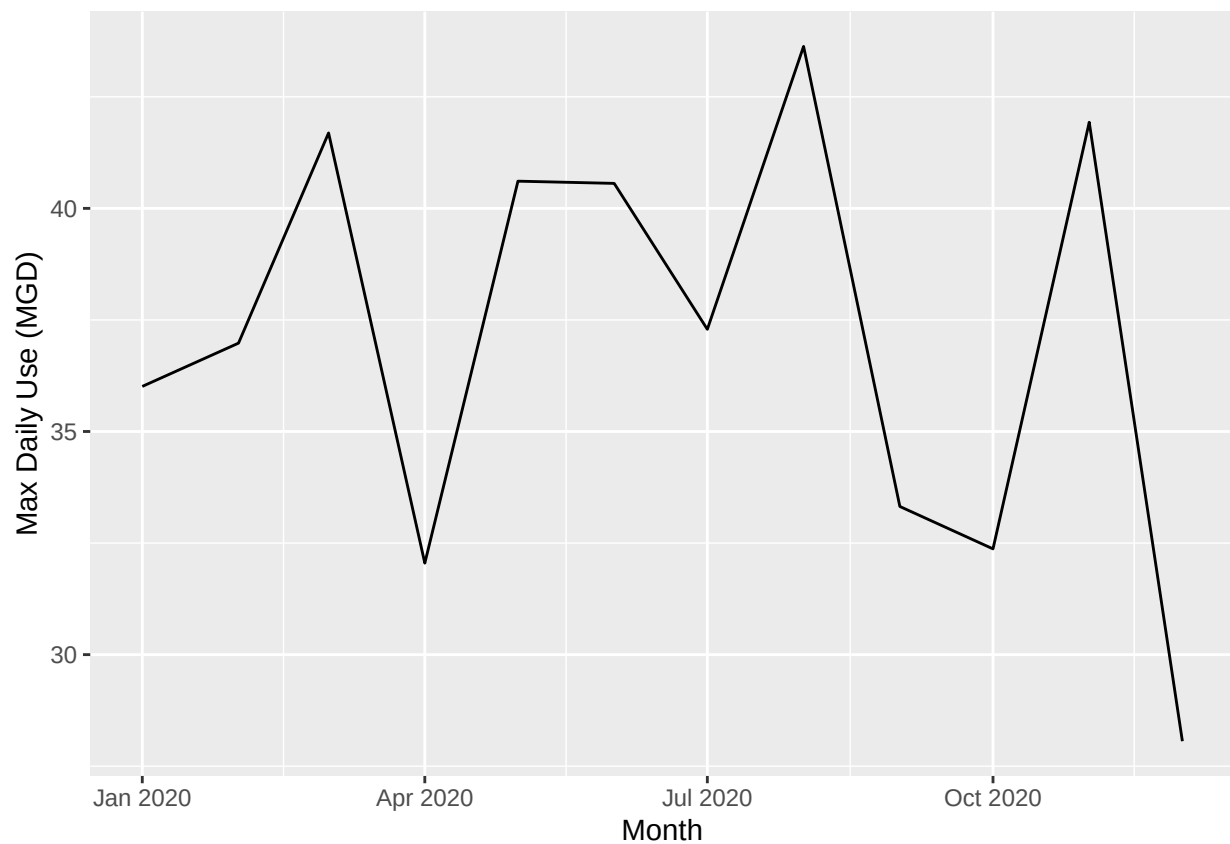
# Fetch the data
df_durham_2020 <- scrape_lwsp("03-32-010", 2020)

# Show structure
str(df_durham_2020)

```

```
## 'data.frame': 12 obs. of 7 variables:
## $ Month      : chr "Jan" "Feb" "Mar" "Apr" ...
## $ Year       : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Max_Day_Use_MGD: num 36 37 41.7 32 40.6 ...
## $ System_Name : chr "Durham" "Durham" "Durham" "Durham" ...
## $ PWSID       : chr "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Ownership   : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Date        : Date, format: "2020-01-01" "2020-02-01" ...
```

```
# Plot maximum daily withdrawals for 2020
ggplot(df_durham_2020, aes(x = Date, y = Max_Day_Use_MGD)) +
  geom_line() +
  labs(x = "Month",
       y = "Max Daily Use (MGD)")
```



- Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8 Fetch and compare Cary and Durham 2020 maximum daily withdrawals

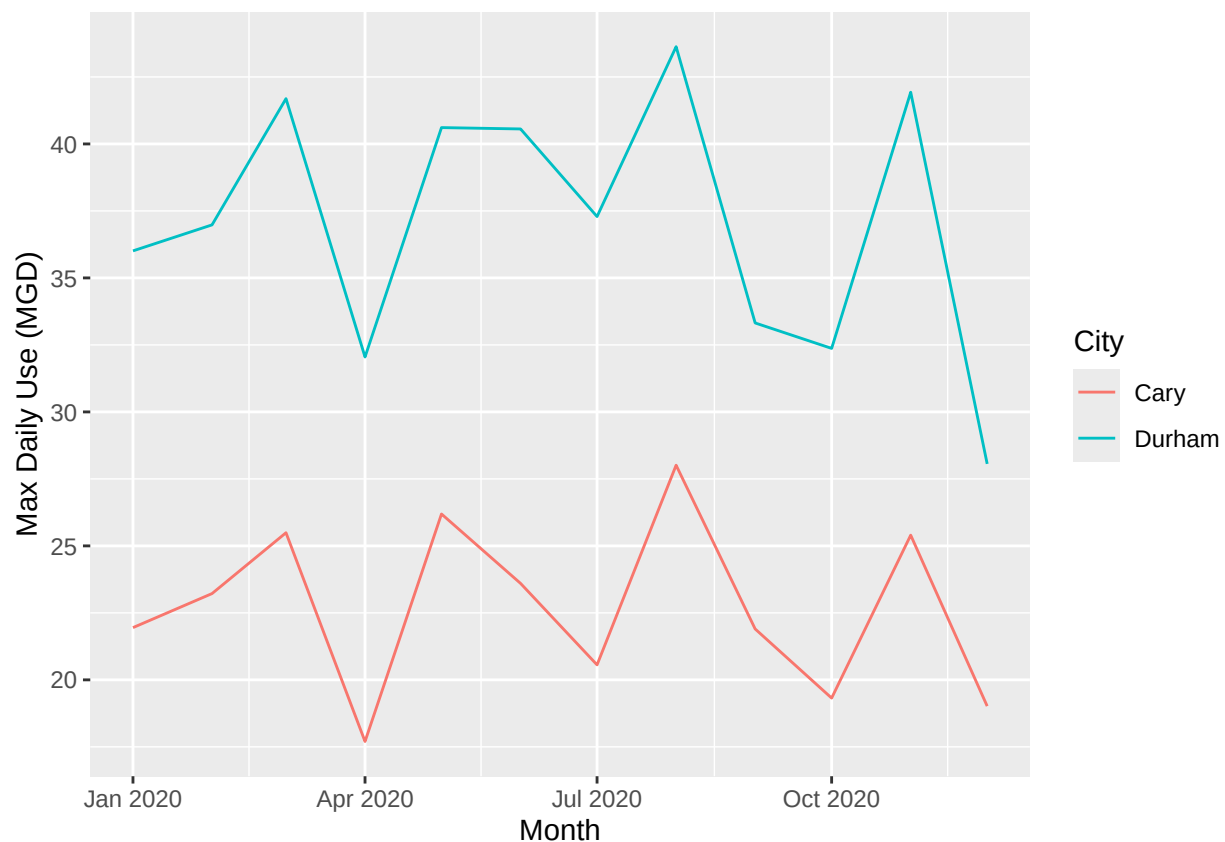
# Durham data
df_durham_2020 <- scrape_lwsp("03-32-010", 2020)

# Cary data
```

```
df_cary_2020 <- scrape_lwsp("03-92-020", 2020)

# Combine into one dataframe for plotting
df_compare_2020 <- bind_rows(
  df_durham_2020 %>% mutate(City = "Durham"),
  df_cary_2020 %>% mutate(City = "Cary")
)

# Plot comparison
ggplot(df_compare_2020, aes(x = Date, y = Max_Day_Use_MGD, color = City)) +
  geom_line() +
  labs(x = "Month",
       y = "Max Daily Use (MGD)")
```



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9 Scrape Cary monthly max daily withdrawal for 2018-2023
```

```
# Define the years of interest
```

```

the_years <- 2018:2023

# Cary PWSID
the_pwsid <- "03-92-020"

# Create a list of PWSIDs the same length as the years vector
the_pwsids <- rep(the_pwsid, length(the_years))

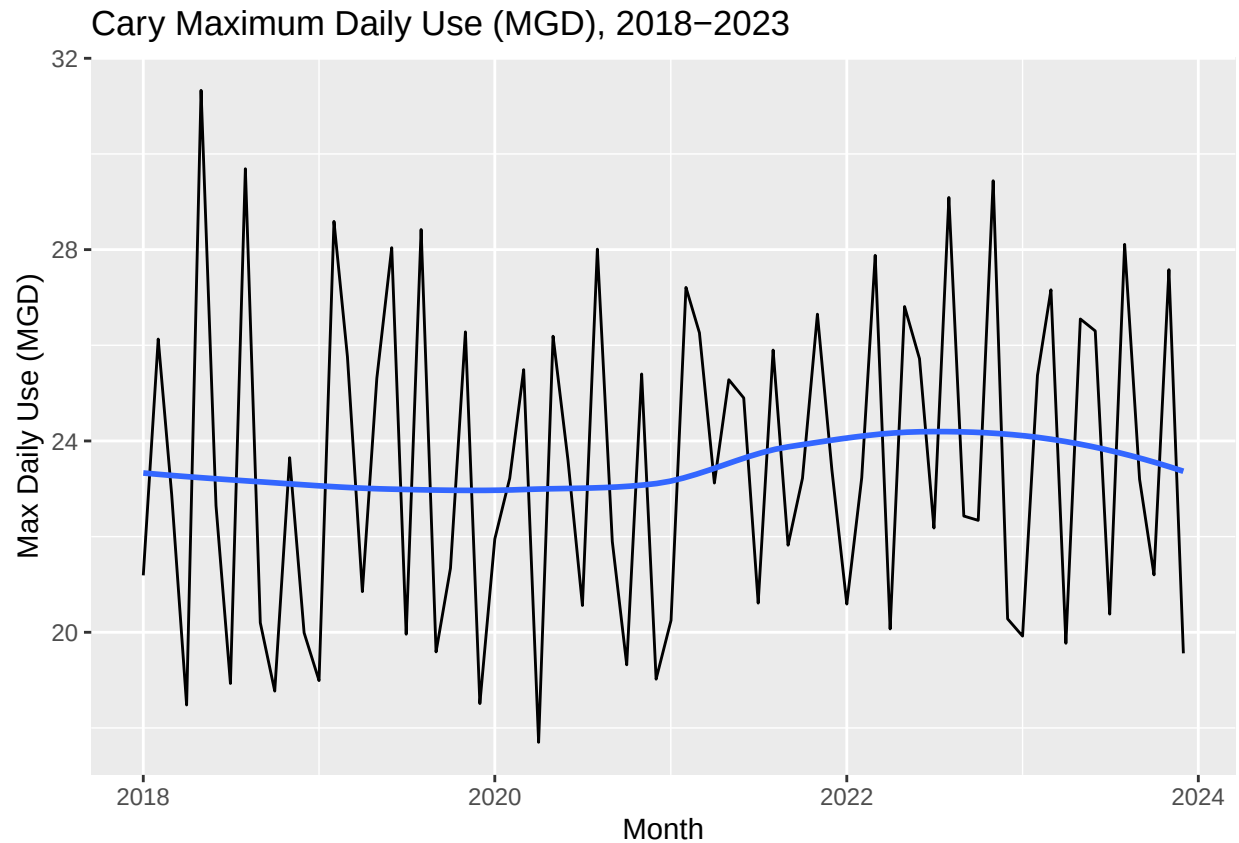
# Use map2
dfs_cary <- map2(
  the_pwsids,
  the_years,
  scrape_lwsp
)

# Combine the list of dataframes into one dataframe
df_cary_all <- bind_rows(dfs_cary)

# Plot monthly maximum daily withdrawals across years
ggplot(df_cary_all, aes(x = Date,
                        y = Max_Day_Use_MGD)) +
  geom_line() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(title = "Cary Maximum Daily Use (MGD), 2018-2023",
       x = "Month",
       y = "Max Daily Use (MGD)")

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: > Based on the plot, Cary does not show a clear upward or downward trend in maximum daily water usage from 2018 to 2023. Although monthly values fluctuate, the overall annual patterns remain relatively stable across the years, and the smoothed loess line is nearly flat.